



Information Technology
Generalist

PROG1400 – Midterm Practical Instructions

Course: PROG1400 – Introduction to Object-Oriented Programming

Instructor: Davis Boudreau

Evaluation Type: Midterm Practical (Hands-On)

Duration: 2 hours (in-class lab)

Purpose of This Midterm

This midterm evaluates your ability to **apply object-oriented programming concepts in practice**, not just describe them.

You will demonstrate that you can:

- Model a game using **UML class and sequence diagrams**
- Use **Mermaid diagrams inside VS Code**
- Use **Copilot responsibly** with proper prompt engineering
- Write **basic object-oriented Python code**
- Manage your work using **GitHub version control**
- Troubleshoot issues and explain your decisions

This midterm is based on **your Pac-Man-like game project** that you selected earlier in the course.

Learning Outcomes Being Assessed

This midterm assesses the following learning outcomes:

- **LO2:** Implement object-oriented design principles in applications
 - **LO3:** Manage and distribute code through code reuse and versioning
 - **LO4:** Develop an object-oriented solution using software modelling documentation
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Allowed Tools

You **will use**:

- VS Code
- GitHub (your repository)

- Mermaid diagrams
- GitHub Copilot
- Course notes and documentation

You **may NOT**:

- Copy code or diagrams from another student
- Blindly accept Copilot output without understanding it

You must be able to **explain everything you submit**.

What You Must Complete

You are required to complete **all three tasks** below. The rubric will be applied to each task.

Task 1 – UML Modelling with Mermaid (LO4)

What You Must Do

1. Open your game project in **VS Code**
2. Ensure a **Mermaid preview extension** is installed and working
3. In your repository, create the folder:

```
docs/uml/
```

4. Create **two Mermaid diagrams**:
 - **One UML Class Diagram**
 - **One UML Sequence Diagram**

Diagram Requirements

Class Diagram

- Represents your game
- Includes **3–5 classes**
- Shows attributes and methods
- Shows relationships between classes

Sequence Diagram

- Shows a real interaction from your game (e.g., movement, collision, scoring, enemy interaction)

Copilot Requirement

- You may use Copilot to help generate Mermaid syntax
- You must **edit, improve, and correct** the output

- Be prepared to explain:
 - What the diagram shows
 - What changes you made
 - Why the diagram represents your game correctly
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Task 2 – Version Control & GitHub (LO3)

What You Must Do

1. Work entirely inside a **GitHub repository**
2. Make **multiple commits** as you work
3. Use **clear commit messages** that describe what you did

You Should Demonstrate

- Understanding of:
 - Local vs remote repositories
 - Why commits are made incrementally
- Ability to explain:
 - What each commit represents
 - Why version control is important

A single large commit at the end will negatively affect your evaluation.

Task 3 – OOP Implementation from UML (LO2)

What You Must Do

1. Using the instructor provided UML Class Diagram, Write **simple Python OOP code** that demonstrates:
 - Classes
 - Encapsulation
 - Inheritance
 - Polymorphism

Important Notes

- Your code does **not** need to be a fully working program.
 - Focus on:
 - Clear class structure
 - Correct responsibilities
 - Be prepared to explain:
 - How your code matches the UML
 - Which OOP principles are being used
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What Will Be Evaluated

You will be evaluated on:

- How well you **manage your time**
- How you **solve problems and troubleshoot**
- How you **use resources**, including Copilot, Personal Notes, Course Handouts
- Your **understanding of OOP and UML concepts**
- Your **hands-on skills** with:
 - VS Code
 - GitHub
 - Mermaid
 - Python

This midterm evaluates **process and understanding**, not perfection.

Tips for Success

- ✓ Think before you code
- ✓ Make small, meaningful commits
- ✓ Use Copilot as a helper, not a replacement
- ✓ Keep UML diagrams simple and accurate
- ✓ Be ready to explain your decisions

Common Mistakes to Avoid

- ✗ Waiting too long to commit
- ✗ Accepting Copilot output without understanding it
- ✗ UML diagrams that don't match your game
- ✗ One class doing everything
- ✗ Not being able to explain your own work

What to Submit

By the end of the midterm, your repository should include:

- `docs/uml/` folder with Mermaid diagrams
- Python source files demonstrating OOP
- A clear commit history showing your work

No separate upload is required unless instructed — your **GitHub repository is your submission**.

Evidence Collected

- Mermaid UML diagrams (`.md`)
 - GitHub commit history
 - Python source files
 - Verbal explanations during the practical
 - Instructor observation of workflow
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Midterm Practical Rubric (Analytic – Brightspace Format)

Scoring Scale: 5 = Exceeded Expectations 4 = Meets Expectations 3 = Approaching Expectations 1 = Does Not Meet Expectations 0 = Not Attempted

Midterm Practical Evaluation Rubric

Criteria	Exceeded Expectations (5)	Meets Expectations (4)	Approaching Expectations (3)	Does Not Meet Expectations (1)	Not Attempted (0)	Score
Time Management <i>(Preparation, organization, task completion)</i>	Excellent preparedness; tasks completed early and efficiently	Well prepared; tasks completed on time	Some preparation; tasks completed late	Poor preparation; tasks incomplete	Task not attempted	/5
Problem Solving Skills <i>(Troubleshooting & application of knowledge)</i>	Independently troubleshoots and applies solutions effectively	Solves most issues with minimal assistance	Requires guidance to resolve issues	Struggles to resolve problems	Task not attempted	/5
Accessing Information Resources <i>(Docs, Copilot, references)</i>	Uses own resources efficiently; finds information very quickly	Uses a mix of resources effectively	Relies on searches but locates information	Struggles to locate or apply information	Task not attempted	/5
Knowledge (Theory Behind the Practical) <i>(UML, OOP principles)</i>	Excellent understanding of UML and OOP concepts; clear explanations	Good understanding with minor gaps	Some understanding; explanations lack depth	Limited understanding	Task not attempted	/5
Hands-On Skills <i>(VS Code, GitHub, Mermaid, Python)</i>	Excellent hands-on skills across all tools	Good hands-on skills	Some hands-on ability	Limited hands-on skills	Task not attempted	/5

Total Midterm Score: /25